Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18RA51]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

CNC MACHINE AND METROLOGY

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Explain the salient features of machining centers and distinguish between vertical machining center and horizontal machining center. **06**

   b. Differentiate between NC machine and CNC machine. **04**

   C Explain the types of CNC and DNC machine. Mention their advantages and disadvantages. **04**

2. a. Explain the importance of spindle drives and axis drives units. **08**

   b. Discuss the operating principles of a Servomotor. Mention its advantages and disadvantages. **06**

3. a. Explain briefly Geometry and wear offset. **05**

   b. Write the manual part program for machining following component on CNC lathe. **09**

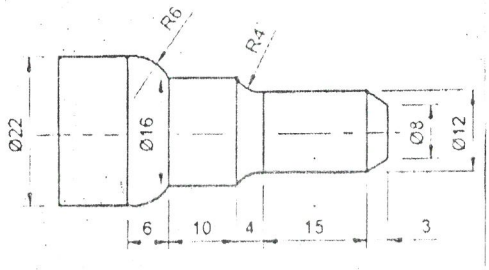

Fig:3b

OR

4. a. Explain the purpose of home position and rapid transverse code and its applications. **07**

   b. Describe interpolation. Explain the purpose of linear and circular interpolation. **07**

5. a. Explain the concept of universal interchange ability and selective assembly **08**

   b. Explain the working principle of sine bar with a neat sketch **06**

OR

6. a. Briefly Explain Go and NO-GO gauge **07**

	b.	Discuss the hole and shaft basic system with neat sketches.	07
7.	a.	Explain the principles of interference and its different types.	07
	b.	Construct and explain the working principle of CMM.	07
		OR	
8.	a.	State the principle involved in the laser.	04
	b.	Explain the construction and working principle of a Michelson Interferometer.	10